



General Characteristics

161. Which one of the following properties is not of transition elements
 (a) Colour
 (b) Paramagnetism
 (c) Fixed valency
 (d) None of the above
162. In which of the following, tendency towards formation of coloured ions is maximum
 (a) s-block elements
 (b) d-block elements
 (c) p-block elements
 (d) f-block elements
163. The electronic configuration $1s^2, 2s^2p^6, 3s^2p^6d^6$ corresponds to
 (a) Mn^{2+} (b) Fe^{2+}
 (c) Co (d) Ge
164. Which of the following statements is not correct about the electronic configuration of gaseous chromium atom
 (a) It has 5 electrons in 3d and one electron in 4s orbitals
 (b) The principal quantum numbers of its valence electrons are 3 and 4
 (c) It has 6 electrons in 3d orbital
 (d) Its valence electrons have quantum number 'l' 0 and 2
165. Zn and Hg belong to the same group, they differ in many of their properties. The property that is shared by both is
 (a) They form oxide readily
 (b) They react with steam readily
 (c) They react with hot concentrated sulphuric acid
 (d) They react with hot sodium hydroxide
166. Which of the following ionic species will impart colour to an aqueous solution
 (a) Ti^{4+} (b) Cu^+
 (c) Zn^{2+} (d) Cr^{3+}
167. The number of electrons in the outermost shell of the 3d-transition elements generally remains
 (a) $(n-1)d^n$ (b) nd^n
 (c) ns^2 (d) $(n-1)s^2$
168. The 3d-elements show variable oxidation states. What is the maximum oxidation state shown by the element Mn
 (a) + 4 (b) + 5
 (c) + 6 (d) + 7
169. Which of the following ions gives coloured solution
 (a) Cu^+ (b) Zn^{++}
 (c) Ag^+ (d) Fe^{++}
170. Which metal represents more than one oxidation state
 (a) Al (b) Na



- (c) *Mg* (d) *Fe*
171. A reduction in atomic size with increase in atomic number is a characteristic of elements of
(a) High atomic masses
(b) *d*-block
(c) *f*-block
(d) Radioactive series
172. Which one of the following characteristics of the transition metals is associated with their catalytic activity
(a) Variable oxidation states
(b) High enthalpy of atomization
(c) Paramagnetic behaviour
(d) Colour of hydrated ions
173. The number of oxidation states of manganese is
(a) 4 (b) 6
(c) 3 (d) 8
174. Which of the following transitional metal has lowest density
(a) *Sc* (b) *Ti*
(c) *V* (d) *Cr*
175. Which of the following transitional metal has lowest boiling point (B.P.)
(a) *Zn* (b) *Sc*
(c) *Ti* (d) *V*
176. Which of the following electronic configuration is that of a transitional element
(a) $1s^2, 2s^2p^6, 3s^2p^6d^{10}, 4s^2p^6$
(b) $1s^2, 2s^2p^6, 3s^2p^6d^{10}, 4s^2p^1$
(c) $1s^2, 2s^2p^6, 3s^2p^6d^2, 4s^2$
(d) $1s^2, 2s^2p^6, 3s^2p^6, 4s^2$
177. *Fe* has been placed in the eighth group, the number of electrons in the outermost orbit is
(a) 1 (b) 2
(c) 3 (d) 4
178. Due to covalent bonding, the transitional metals are
(a) Lustrous (b) Conductor
(c) Hard and brittle (d) Ductile
179. The magnetic moment of a metal ion of first transition series is 2.83 *BM*. Therefore it will have unpaired electrons
(a) 6 (b) 4
(c) 3 (d) 2
180. Which of the following pair of ions may exhibit same colour
(a) Cr^{+++} and Fe^{++}
(b) Ti^{+++} and V^{++}
(c) Fe^{+++} and Mn^{++}
(d) Cu^+ and Ni^{++}

